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Introduction to Glomerular Diseases

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Biomedical Sciences
&
Former Academic Medicine scholar, Dr. Jerry Jose

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Learning Objectives

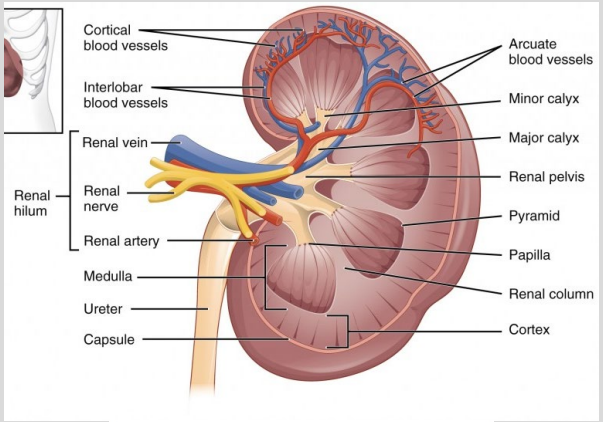
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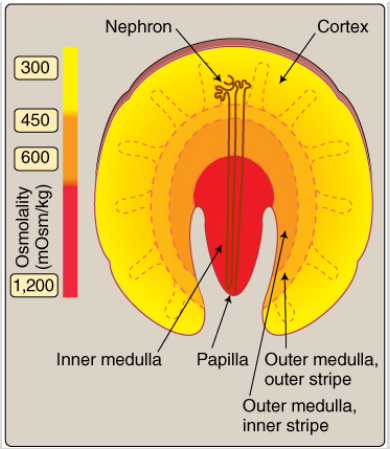
Learning Objectives By the end of this introductory session, students should be able to:

1. Describe the basic structure and function of the glomerulus within the nephron
2. Explain how disruptions in the glomerular filtration barrier lead to clinical disease
3. Understand general mechanisms that contribute to glomerular injury (immune complex deposition, antibody- and cell-mediated damage, complement activation)
4. Differentiate broadly between nephrotic and nephritic syndromes in terms of pathophysiology and clinical features
5. Appreciate the role of diagnostic tools such as light microscopy, immunofluorescence, and electron microscopy in evaluating glomerular pathology
6. Recognize hallmark features seen in common glomerular diseases without going into subtype-level details

Kidney Anatomy Overview

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The kidney's gross anatomy **Hilum:** Entry/exit site for renal artery, vein, lymphatics, and ureter

- **Capsule:** Outer protective fibrous covering of the kidney
- **Cortex:** Contains **all renal corpuscles** (glomeruli + Bowman's capsules), and is the site of **initial blood filtration**
- **Medulla:** Contains loops of Henle and collecting ducts, involved in **urine concentration and tubular processing**
- **Minor and Major Calyces → Renal Pelvis:** Collect urine from the papillae and funnel it into the ureter
- **Blood Flow Pathway:** Renal artery → segmental → interlobar → arcuate → interlobular → **afferent arteriole → glomerulus** → efferent arteriole

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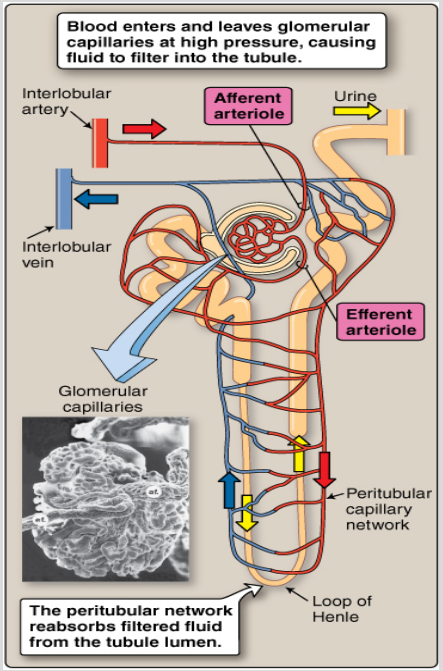
The Nephron

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- Site of initial blood filtration
- Nephron: The functional unit of the kidney

It consist of:

1. **Renal corpuscle**
2. **Renal tubule** (proximal tubule, loop of Henle, distal tubule, collecting duct; Responsible for reabsorption and secretion)
3. **Afferent and efferent arteries**
4. **Interstitium**



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1. Renal corpuscle

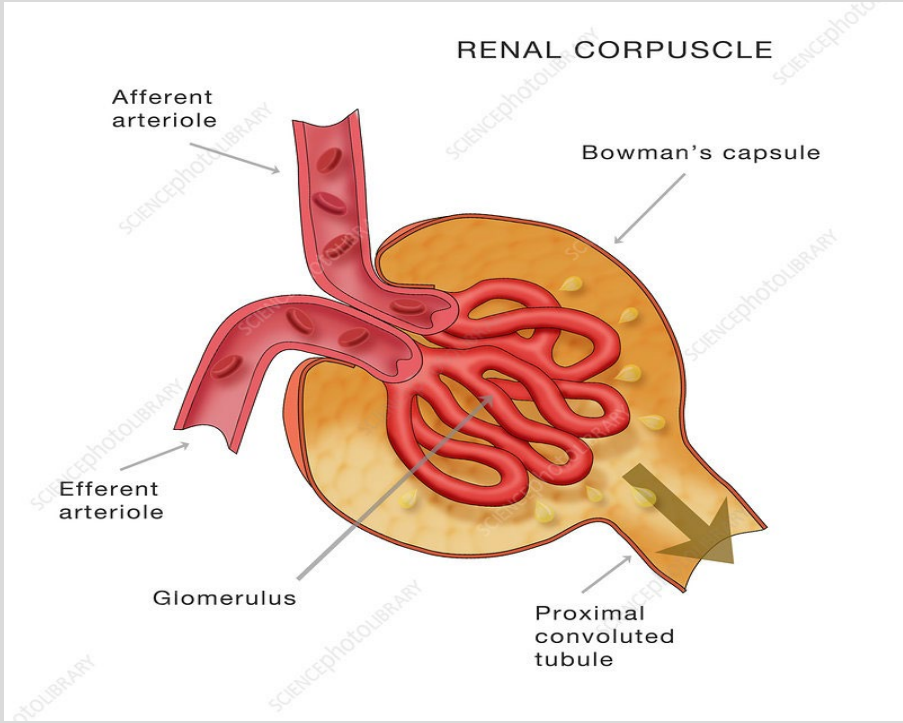
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1.a:Glomerulus

Tuft of capillaries involved in filtration

1.b: Bowman's capsule

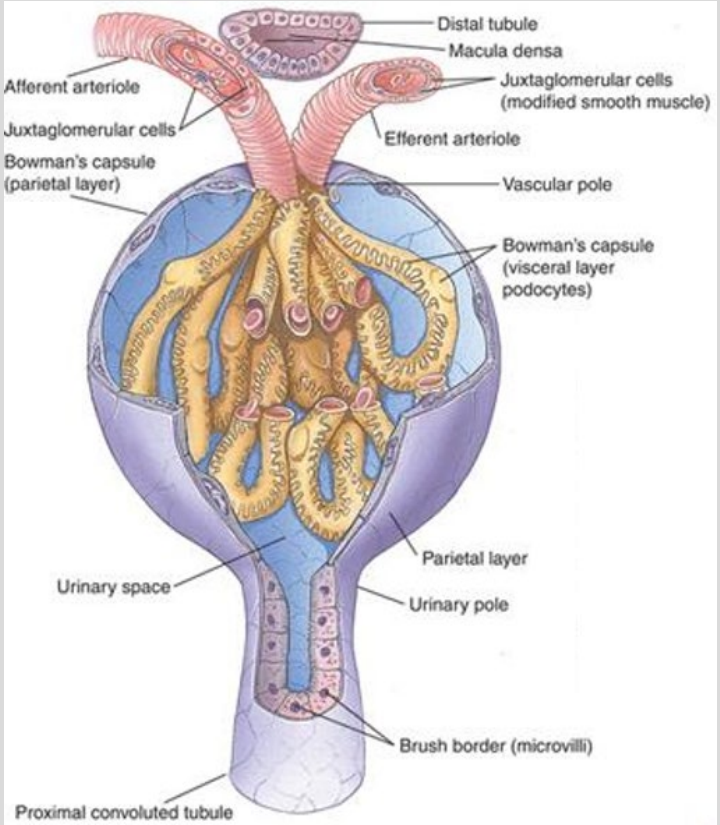
surrounds the glomerulus and collects filtrate

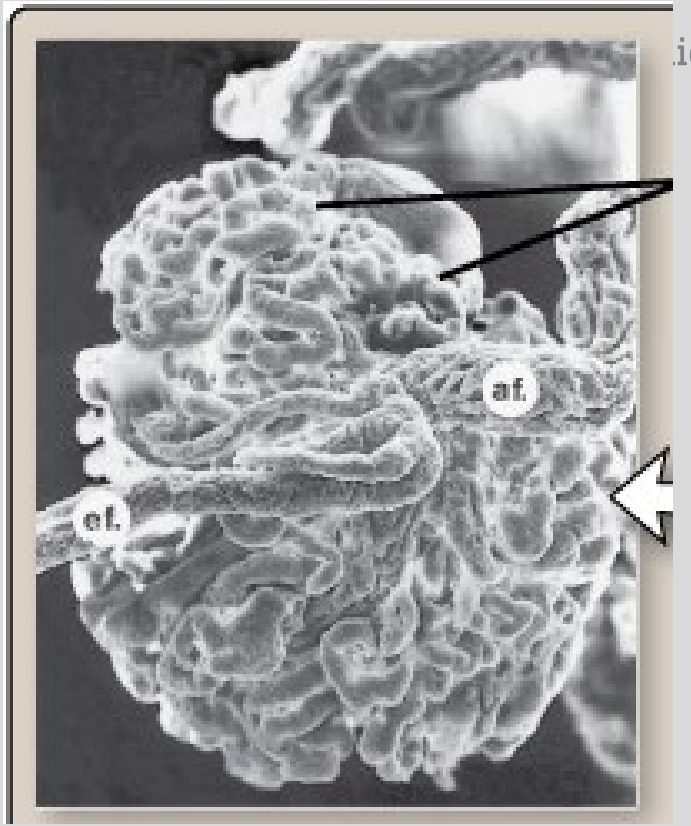


https://www.sciencephoto.com/media/704407/view/renal-corpuscle-illustration

1.a: The Glomerulus

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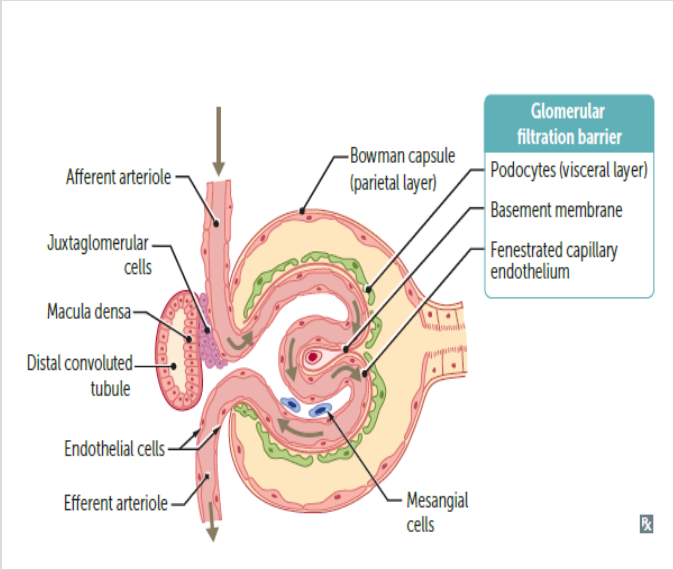


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Gartner, Leslie P. *Color Atlas and Text of Histology 7e ch16*

The Glomerular Filtration Barrier

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- Fenestrated capillary endothelium :
  - Fenestrated (~70nm)
  - Allows water, solutes, and proteins
- Glomerular basement membrane (GBM) - negatively charged:
  - Repels negatively charged molecules, e.g. proteins



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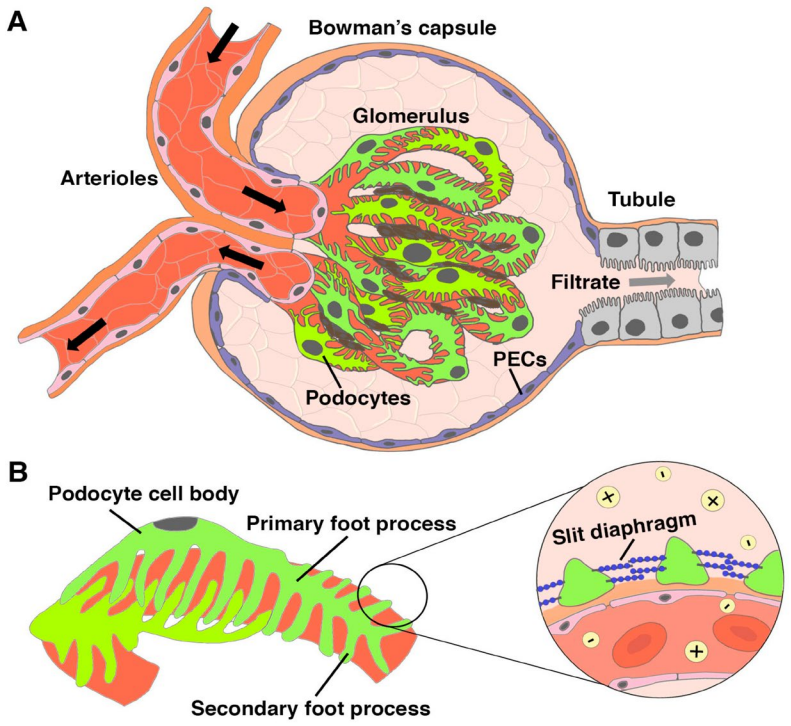
1.b: The Bowman's capsule

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- Podocyte foot processes (slit diaphragm) :

blocks proteins and other large molecules to pass through

- Parietal layer (Forms the outer wall of the capsule)

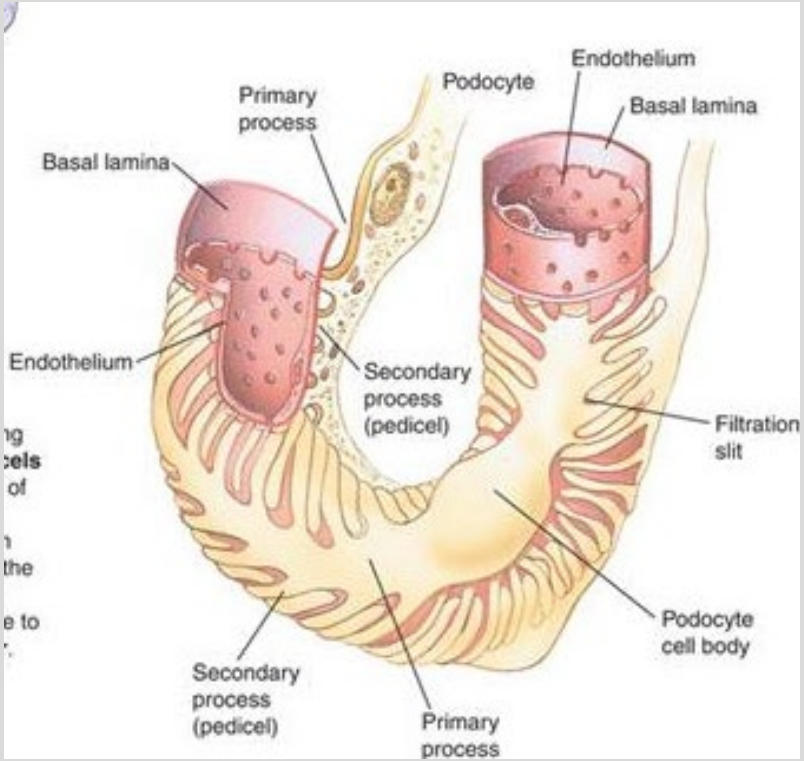


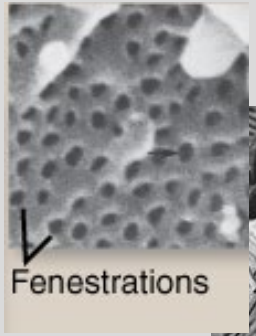
https://images.app.goo.gl/rRYriZYdyRp2FZGs6

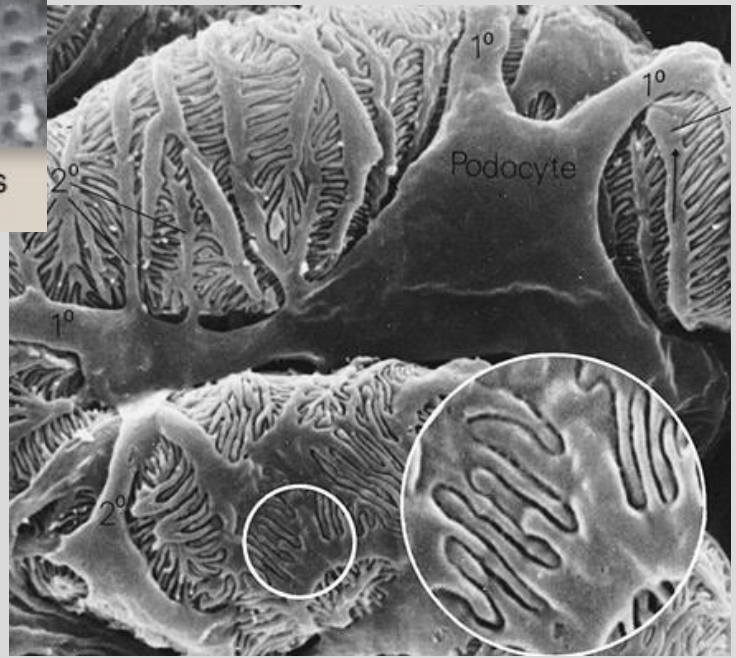
Podocytes and Fenestrated Endothelium

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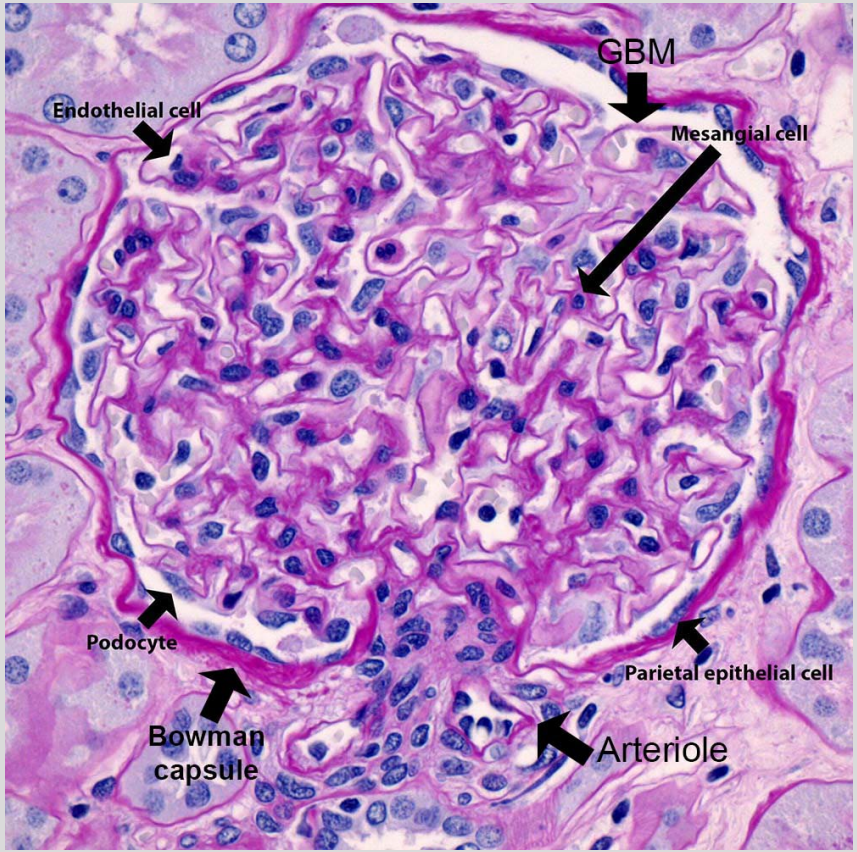




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Gartner, Leslie P. *Color Atlas and Text of Histology 7e ch16*

Glomerular Anatomy

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- Endothelial cells
- GBM
- Podocyte
- Bowman's space
- Parietal epithelia
- Mesangium

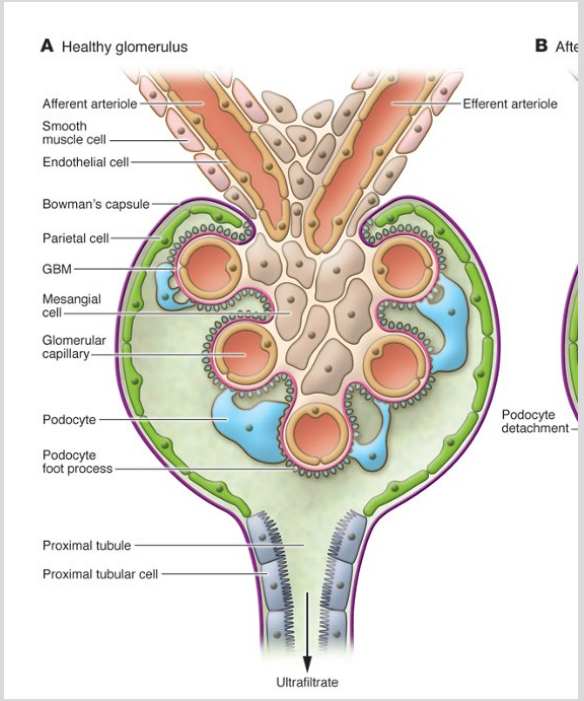
https://www.renalfellow.org/2019/01/04/kidney-biopsy-of-the-month-what-is-normal/

Mesangium

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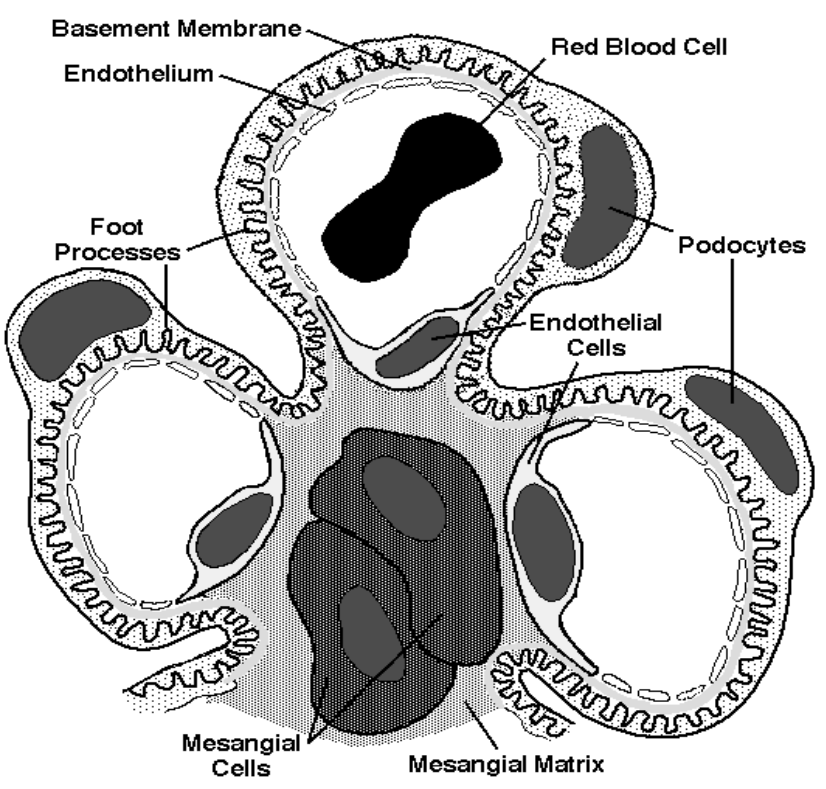
- Central position in the renal glomerulus
- Has the modified smooth muscle cell characteristics
- ~15% of glomerular filtration occurs through the mesangium
- Main function is to support glomerular capillaries
- Contains macrophage-like cells called mesangial cells
  - Can release inflammatory mediators and proliferate (we will see in some of the disease)

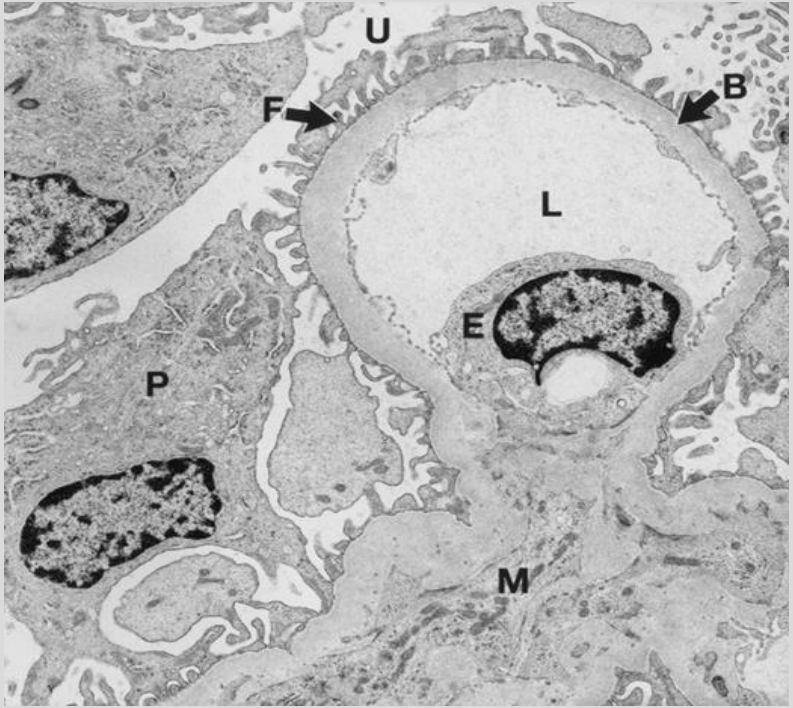


chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/http://content-assets.jci.org/manuscripts/57000/57935/JCI57935.v2.pdf

Capillary loops and the Mesangium

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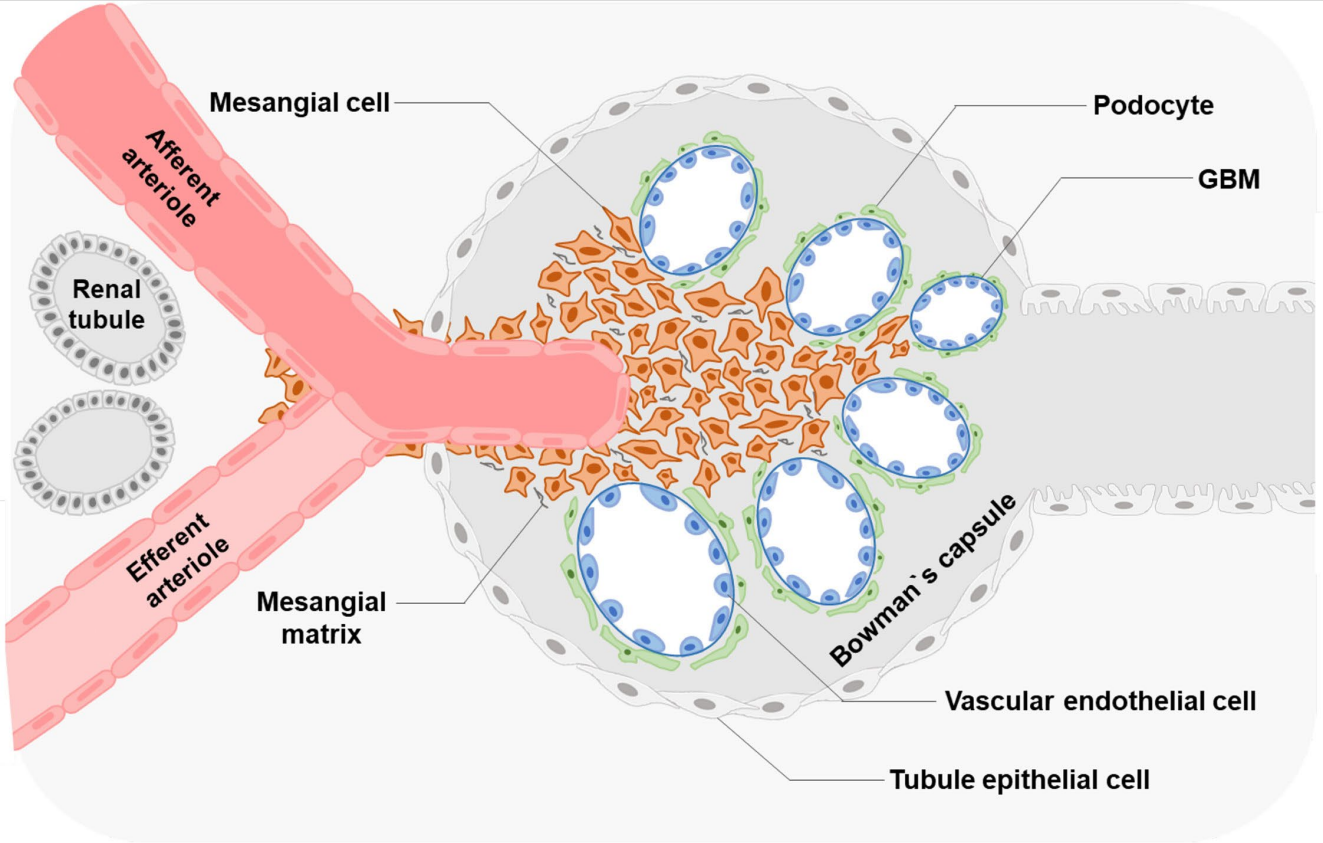


https://medicine.tamu.edu/class-files/webpath16/renahtml/renal109.htm
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All of filtration barrier put together

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https://www.researchgate.net/figure/Glomerular-structure-and-MCs-location-Glomerular-MCs-and-the-mesangial-matrix-form-a_fig1_366293595

Potential causes of damage to Filtration barrier:

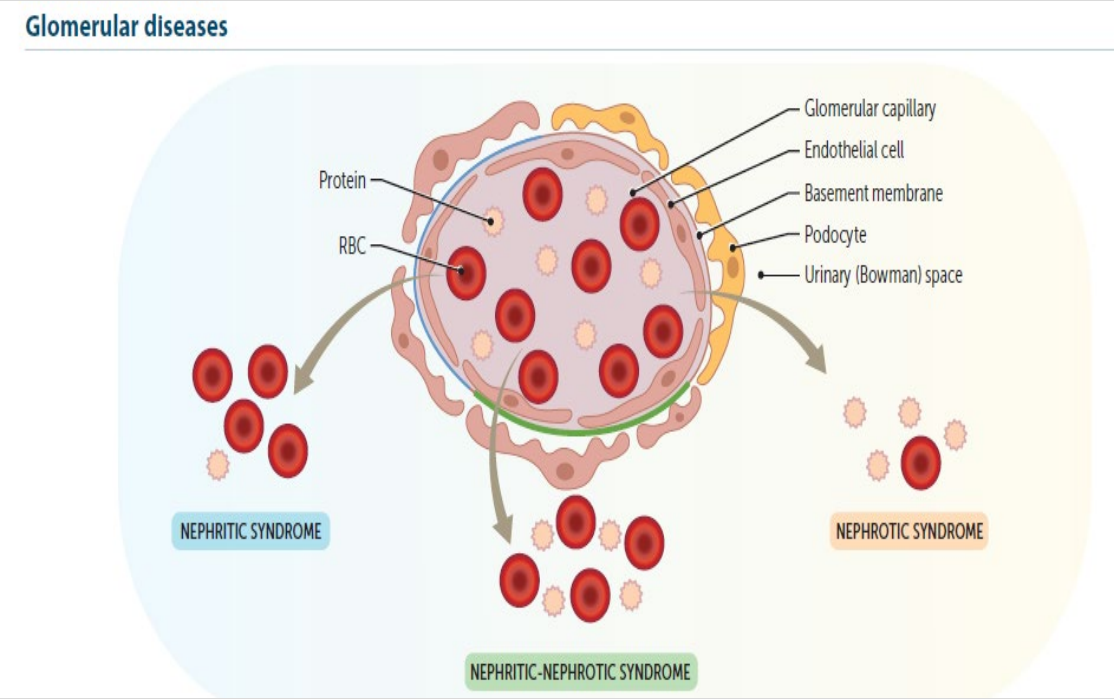
- **Antibody directed destruction**
  - Type 2 hypersensitivity reaction (HSR)
- **Immune complex deposition**
  - Type 3 HSR
- **Cell mediated destruction**
  - Type 4 HSR
- **Complement system activation**
  - Alternative pathway

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https://www-uptodate-com.arktos.nyit.edu/contents/glomerular-disease-evaluation-and-differential-diagnosis-in-adults?search=glomerular%20disease&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

Glomerular Diseases presentation

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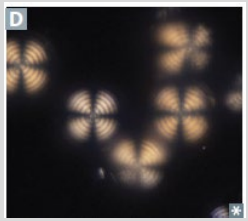
- Damage to glomerular morphology causing disruption of the filtration barrier
- Can cause leakage of either:
  - Protein in urine (proteinuria)
  - Blood in urine (Hematuria)

NephrOtic Syndrome=PrOteinuria

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- Glomerular injury will be due to cytokines, not neutrophils
- Podocytes are impaired and can NOT repel negatively charged proteins → Massive prOteinuria
  (> 3.5 g/day)
- Some special proteins that can be lost:
  Albumin (Albuminuria)
  Ig G Antibody (increased risk of infection, and thus hypogammaglobulinemia)
  Anti-thrombin-III (Hypercoagulable)
- The liver compensating for loss of protein and tries to increase osmolarity by increasing cholesterol production (frothy urine and fatty casts in urine (Maltese Cross sign), and hypercholesteremia)





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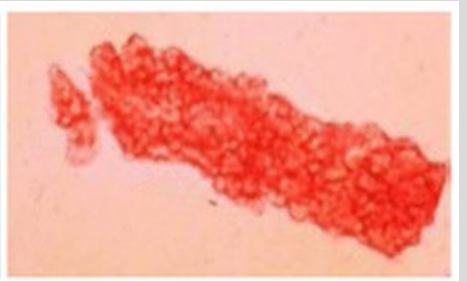
Nephritic Syndrome

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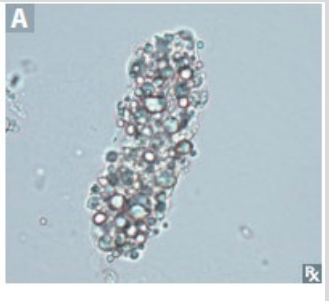
- Glomerular inflammation (cellular proliferation with neutrophilic infiltration)

Clinical presentation:

- GBM damage → loss of RBCs into urine → dysmorphic RBC and RBC casts in the urine
- Decrease in GFR → Oliguria, Azotemia (increased BUN/Cr)
- Renin release increases→ HTN
- Proteinuria (<3.5 g/day) but in severe cases it may reach higher



https://theory.labster.com/casts_url/



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Nephritic and Nephrotic Syndromes

TYPE	ETIOLOGY	CLINICAL PRESENTATION	EXAMPLES
Nephritic syndrome	Glomerular inflammation → GBM damage → loss of RBCs into urine → dysmorphic RBCs, hematuria	Hematuria, RBC casts in urine ↓ GFR → oliguria, azotemia ↑ renin release, HTN Proteinuria often in the subnephrotic range (< 3.5 g/day) but in severe cases may be in nephrotic range	▪ Infection-associated glomerulonephritis ▪ Goodpasture syndrome ▪ IgA nephropathy (Berger disease) ▪ Alport syndrome ▪ Membranoproliferative glomerulonephritis
Nephrotic syndrome	Podocyte damage → impaired charge barrier → proteinuria	Massive proteinuria (> 3.5 g/day) with edema, hypoalbuminemia → ↑ hepatic lipogenesis → hypercholesterolemia Frothy urine with fatty casts Associated with hypercoagulable state due to antithrombin III loss in urine and ↑ risk of infection (loss of IgGs in urine and soft tissue compromise by edema)	May be 1° (eg, direct podocyte damage) or 2° (podocyte damage from systemic process): ▪ Focal segmental glomerulosclerosis (1° or 2°) ▪ Minimal change disease (1° or 2°) ▪ Membranous nephropathy (1° or 2°) ▪ Amyloidosis (2°) ▪ Diabetic glomerulonephropathy (2°)

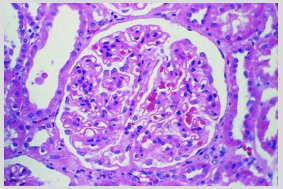
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Diagnostic Techniques for Glomerular Disease:

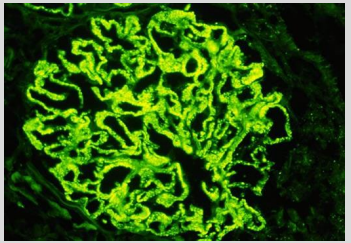
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Diagnosis is guided by clinical presentation and confirmed through renal biopsy analysis

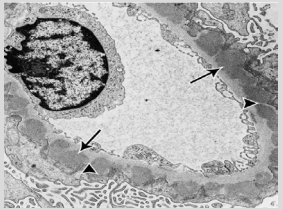
- Light microscopy (LM)
  Gross structure of glomerulus and cellularity



- Immunofluorescence (IF)
  Pattern and type of protein (Ab, complement) deposition



- Electron microscopy (EM)
  Identifies location of protein deposition
  Foot process effacement



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Light Microscopy (LM)

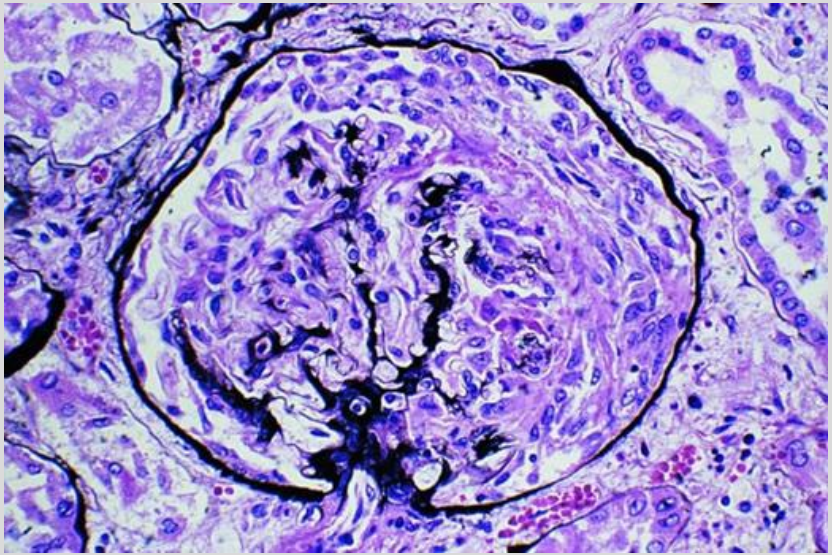
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- **Identifies areas of**
  - Increased cellularity
  - Increased extracellular matrix
- **Stains**
  - H&E: basic morphology
  - PAS: basement membrane & mesangium
  - Silver stain: GBM pattern
  - Congo Red: amyloid
  - Trichrome: fibrosis, sclerosis

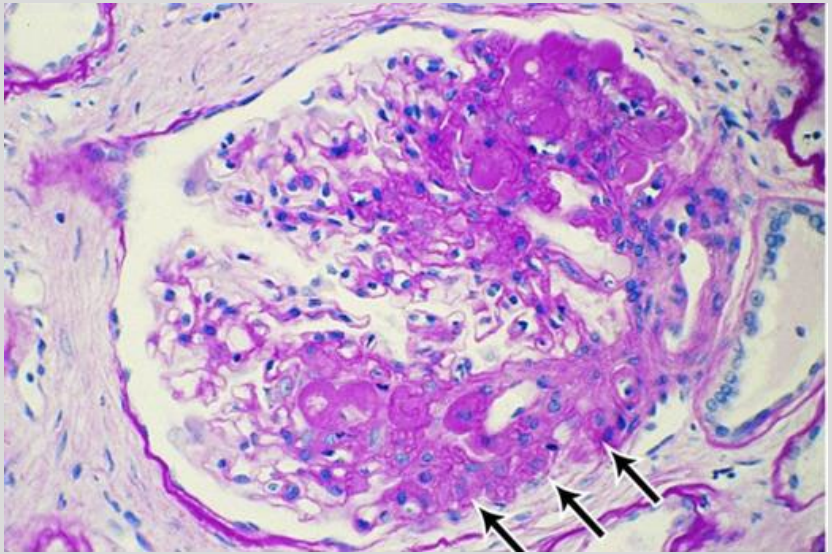
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Light Microscopy

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Silver staining (shows ANCA glomerulonephritis)

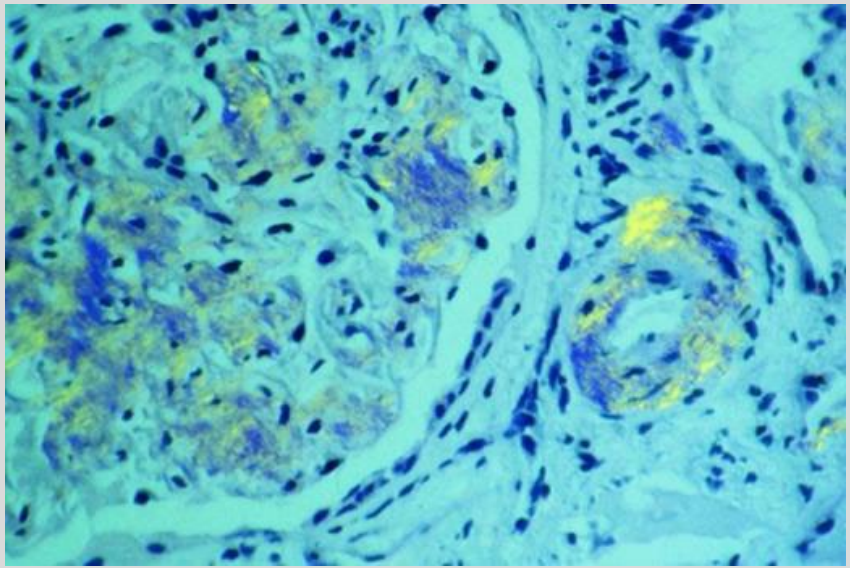


PAS staining (shows FSG)

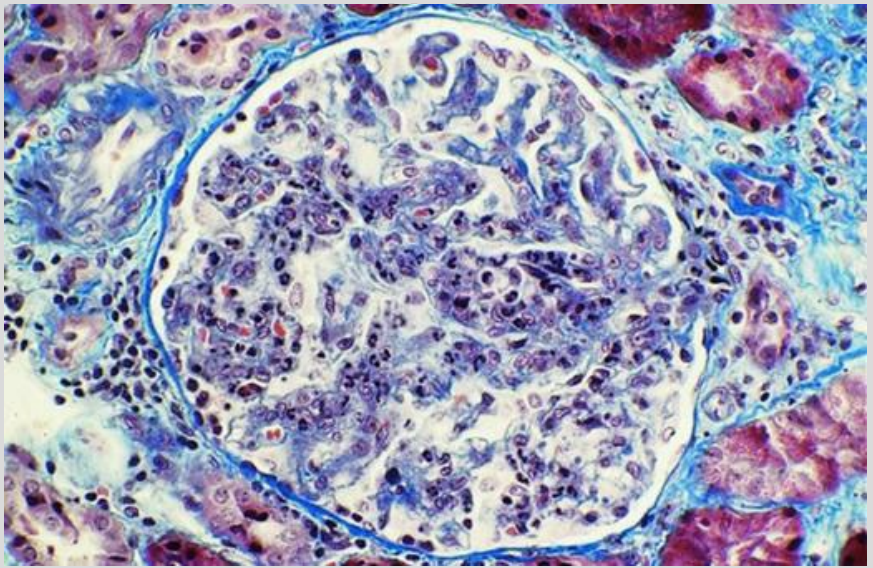
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Light Microscopy

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Congo red staining (shows amyloid nephropathy)



Trichrome staining (shows acute PSGN)

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Immunofluorescence

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- **Identifies patterns of:**

Linear: anti-GBM (Goodpasture)

Granular: immune complexes (PSGN, lupus)

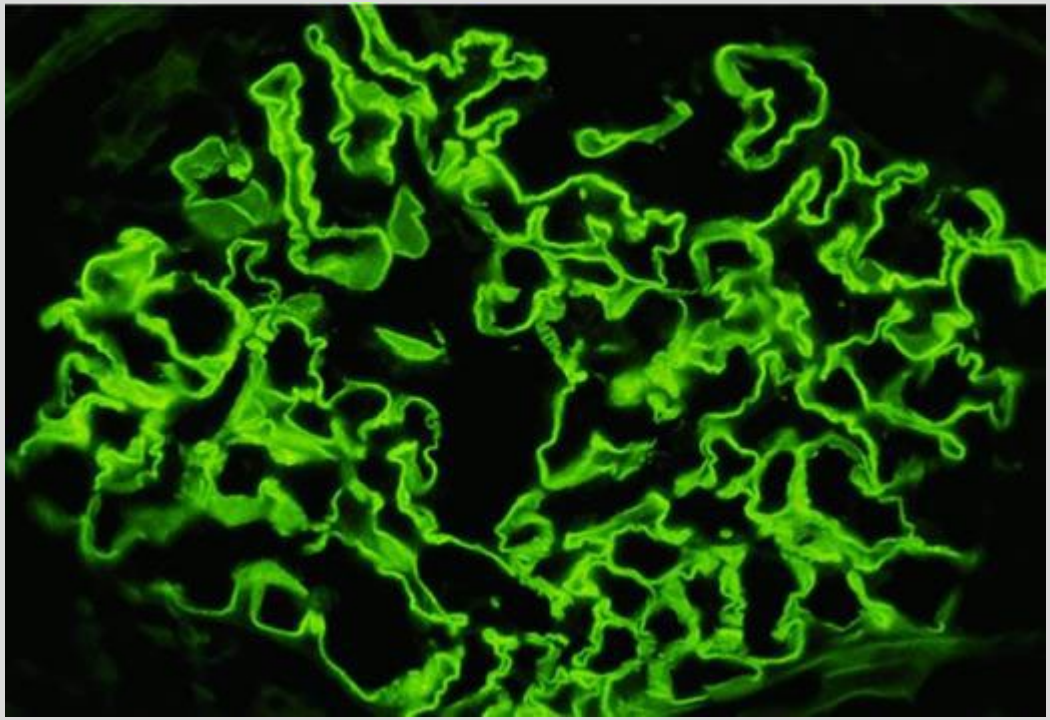
Irregular/amorphous: IgA nephropathy

- **Detects:**
  - Albumin
  - Fibrinogen
  - Antibodies (IgA, IgG, IgM)
  - Complement (C1q, C3, C4)

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Linear Immunofluorescence Staining

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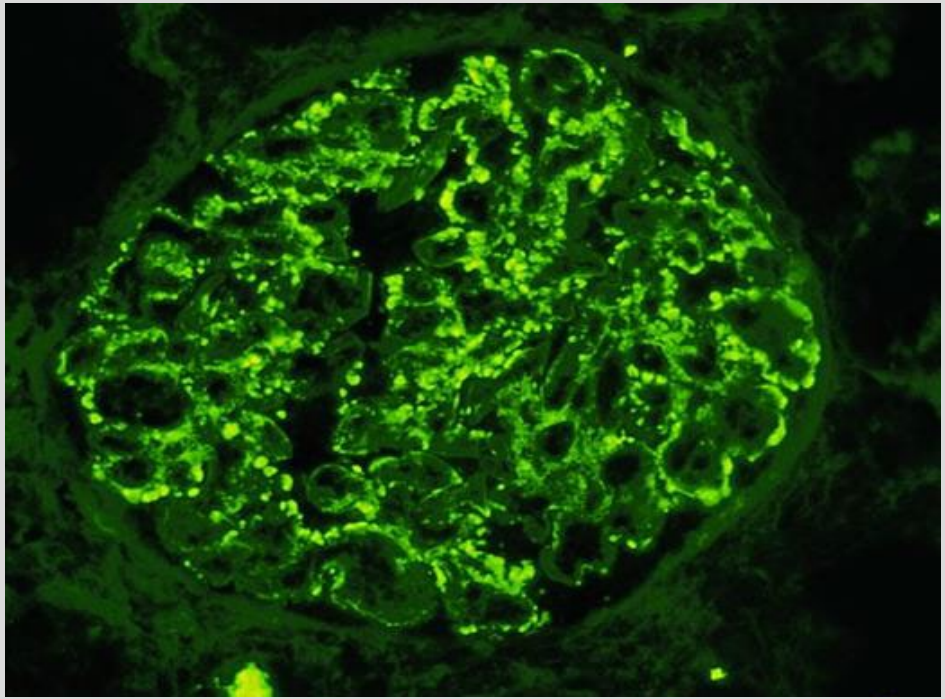


- Immunostaining for IgG
- Diffuse linear pattern
- Seen in anti-GBM disease
  - Type 2 hypersensitivity reaction

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Granular Immunofluorescence Staining

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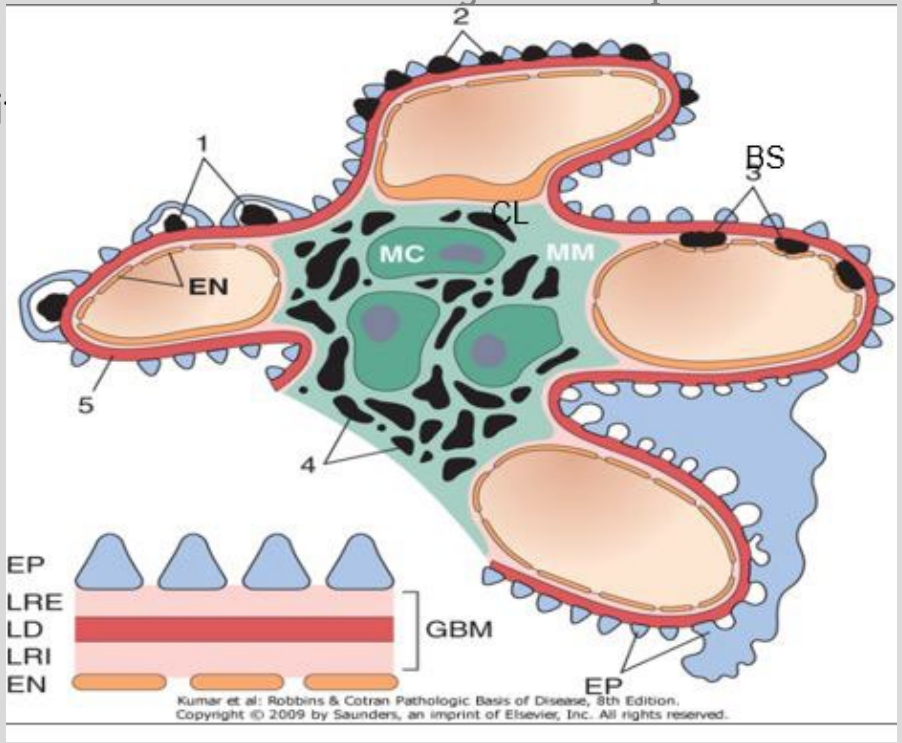
- Immunostaining for immunocomplex deposits
- Diffuse granular pattern
  - "lumpy-bumpy"
- Seen in acute post-streptococcal glomerulonephritis
  - Type 3 hypersensitivity reaction

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Electron Microscopy

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- Identifies:
  - Electron dense immune complex deposi
    - Subepithelial humps/deposits
    - Subendothelial
    - Mesangial
  - GBM thickening and remodeling
  - Collagenous matrix
  - Fibrillar deposits
- Detects:
  - Effacement of foot processes
  - Endothelial injury

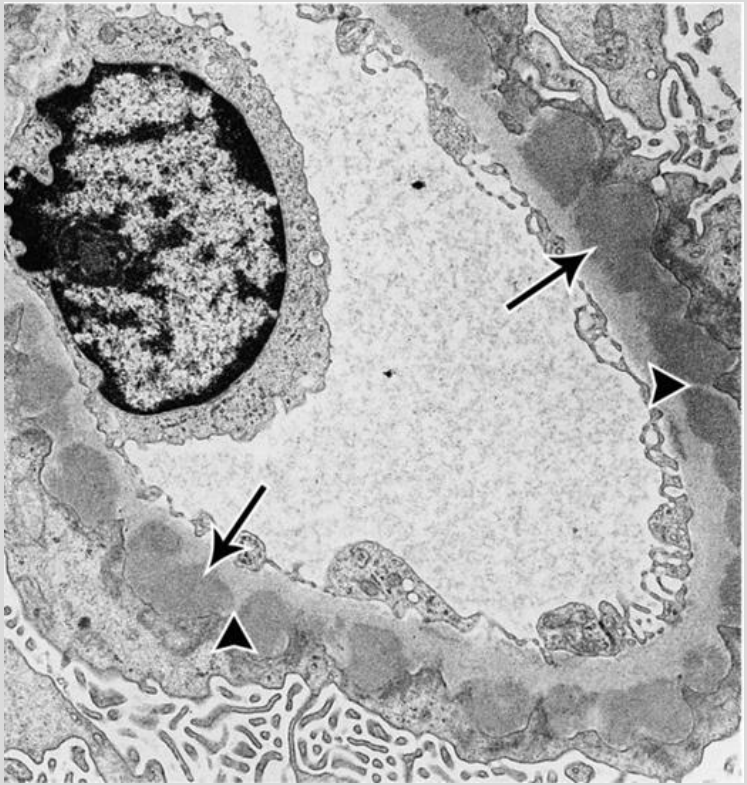


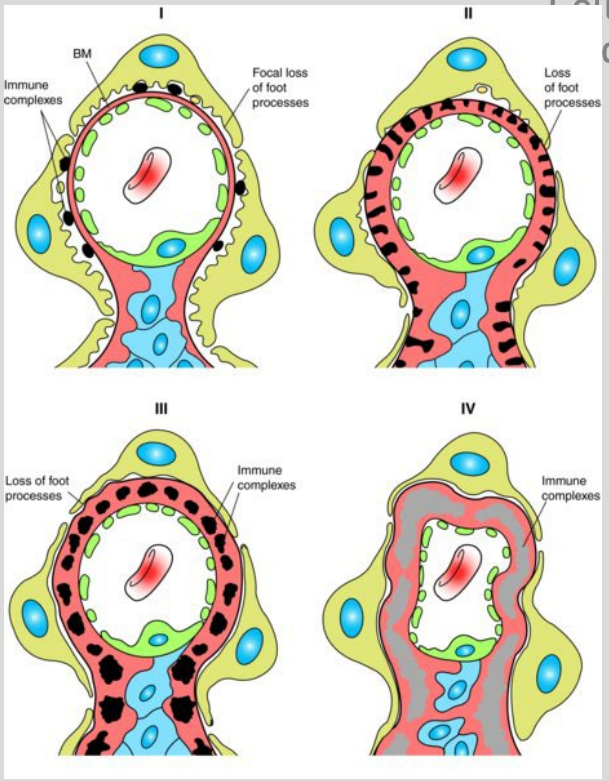
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Vinay, Kumar. *Robbins & Cotran pathological Basis of Disease 8e*

Electron Microscopy (Subepithelial)

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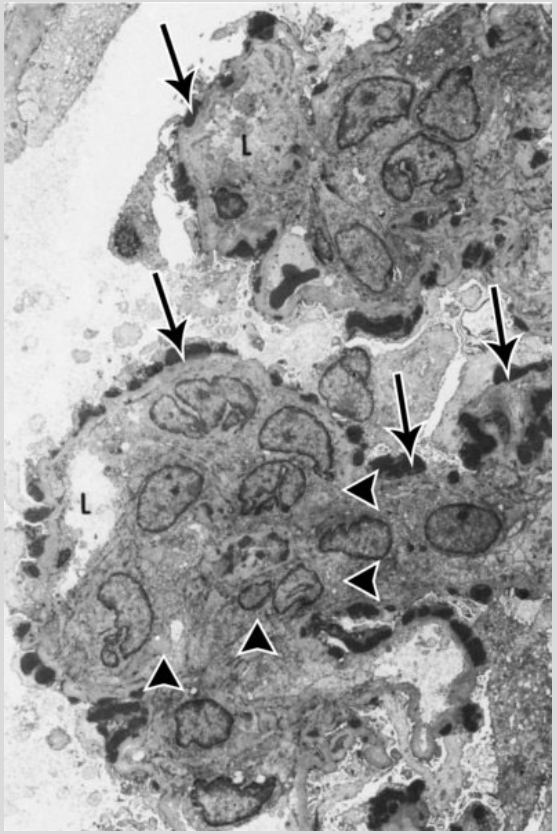


- Membranous nephropathy

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Electron Microscopy (Subepithelial)

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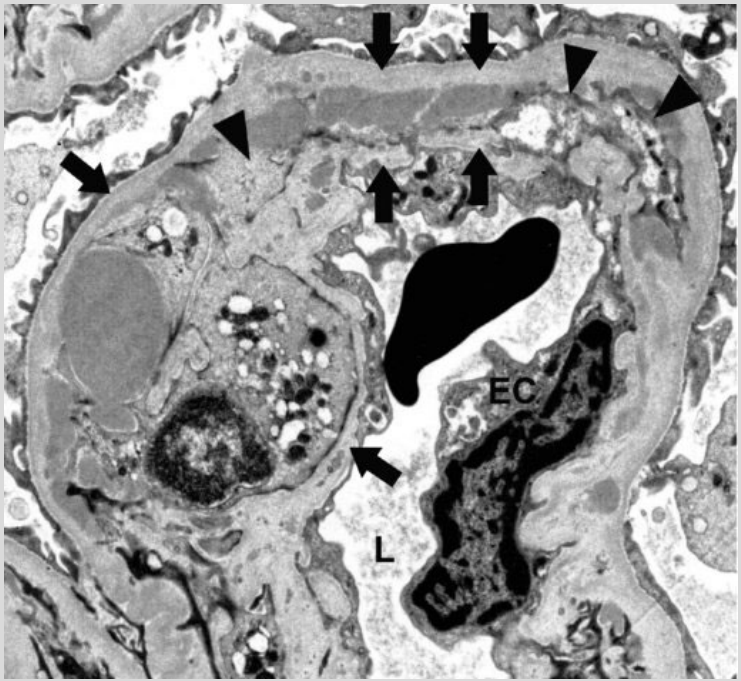


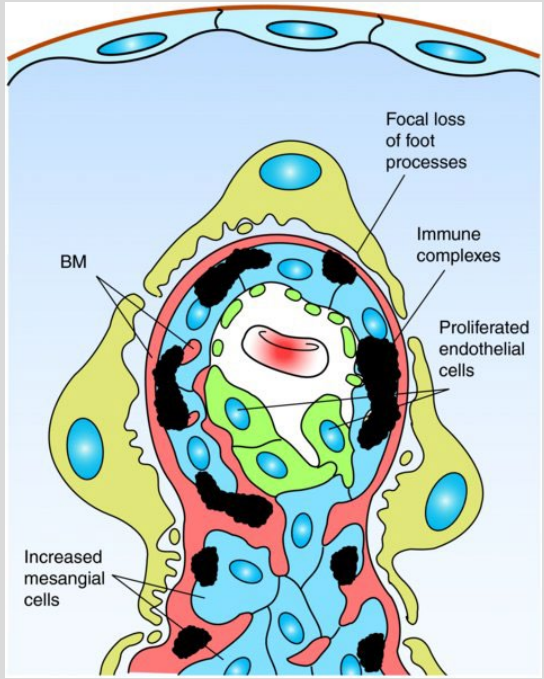
- Acute post-streptococcal glomerulonephritis (PSGN)
- Subepithelial humps
- Mesangial hypercellularity

Strayer, David S., *Rubin's Pathology: Mechanisms of Human Disease 8e ch22*

Electron Microscopy (Subendothelial)

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- Membranoproliferative glomerulonephritis
  - "Tram track" appearance
- Subendothelial deposits with mesangial cell proliferation

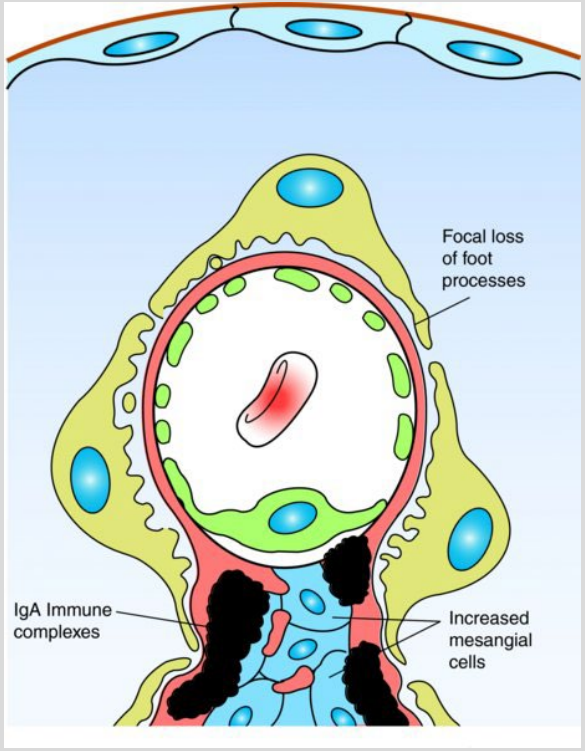
Strayer, David S., *Rubin's Pathology: Mechanisms of Human Disease 8e ch22*

Electron Microscopy (Mesangial)

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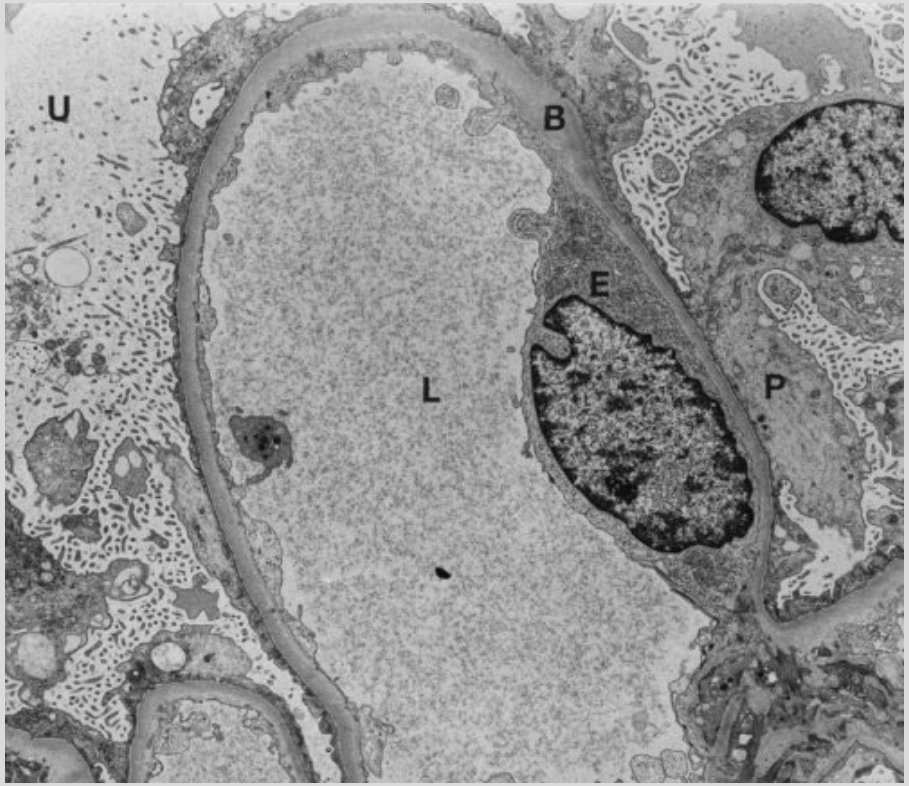


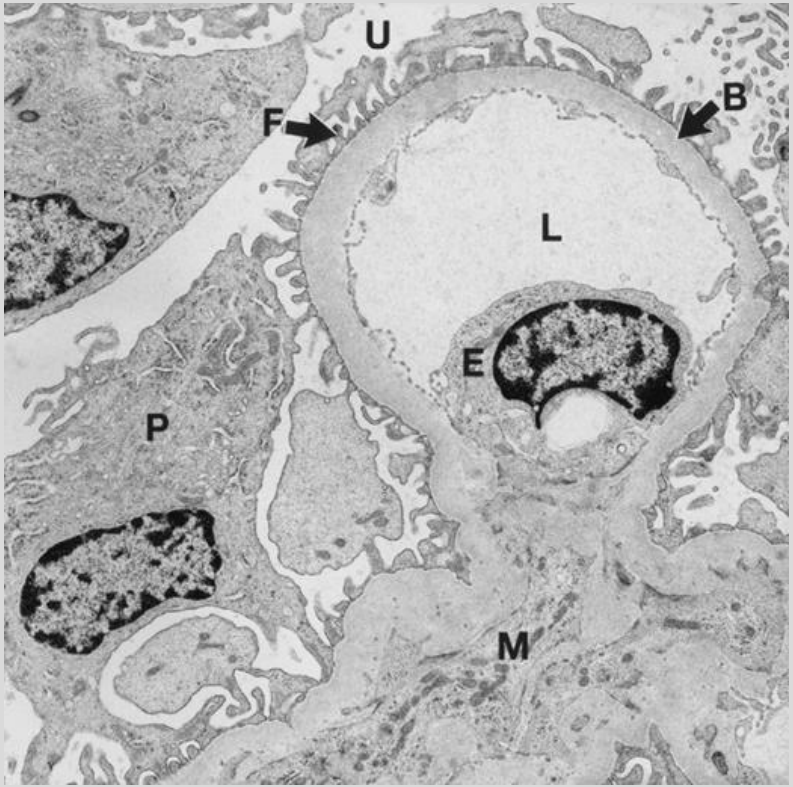
- IgA nephropathy
- Mesangial matrix deposits between mesangial cells and GBM

Strayer, David S., *Rubin's Pathology: Mechanisms of Human Disease 8e ch22*

Effacement of the foot processes

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Summary

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- Filtration barrier: critical to glomerular function
- Glomerular disease = damage to barrier
- Nephrotic syndrome: characterized by proteinurea and hypoalbuminemia (PALE)
- Nephritic syndrome: characterized by inflammation of the glomeruli and resulting hematuria
- Microscopy essential for diagnosis

Sample Question

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- Which of the following findings best differentiates nephritic from nephrotic syndrome on urinalysis?

  A) Presence of hyaline casts
  B) Presence of dysmorphic RBCs and RBC casts
  C) Presence of fatty casts
  D) Low urine sodium

Sample Question

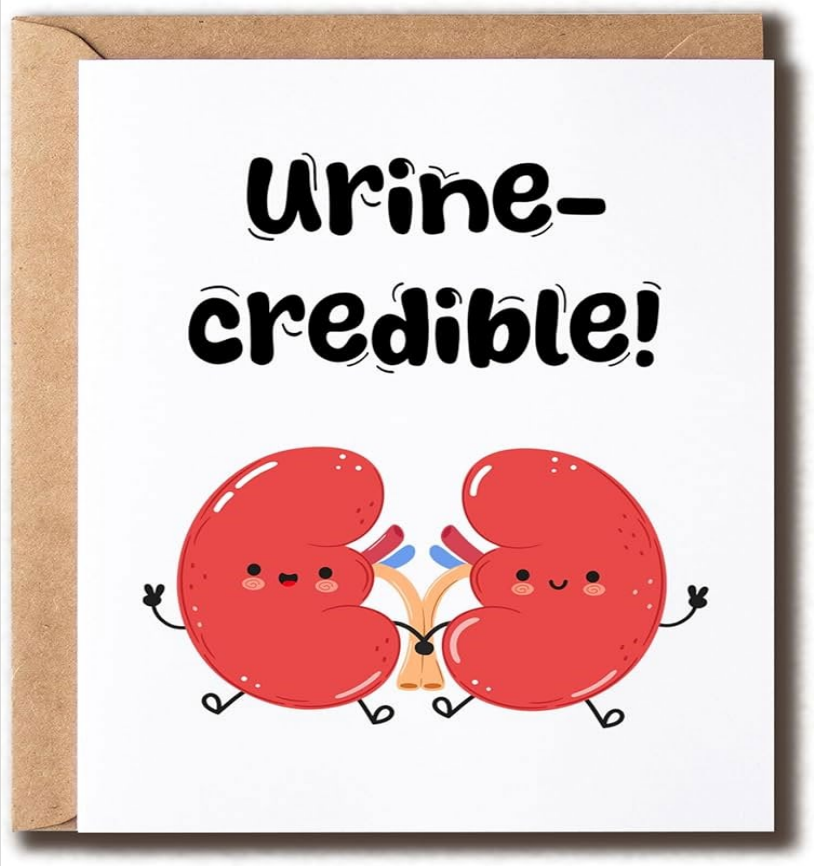
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  C) Presence of fatty casts
  D) Low urine sodium

Explanation:
Nephritic syndrome involves glomerular inflammation, leading to **hematuria** with **dysmorphic red blood cells** and **RBC casts** on urinalysis. This finding is **specific to nephritic syndrome** and helps differentiate it from nephrotic syndrome, which is marked by heavy proteinuria and **fatty casts**.

Thank You!



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Lecture Feedback Form:

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